Two Margins of International Athlete Recruitment to NCAA Division I: Entry, Sublinear Scaling, and Home-Country Shock Response

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Abstract

We construct an international NCAA Division I athlete panel from a 2024–25 cross-section of 19,317 track-and-field and soccer rosters, an eight-sport multi-year panel of 260,737 athlete-years (2019–2023), and a subsample of TFRRS individual season-best marks. **On the extensive margin**, log GDP per capita and log population predict whether a country sends any athletes ($\hat{\beta} = 0.52$, 0.18; both $p < 0.01$); political stability does not. **On the intensive margin**, athletes scale sublinearly with source-country population: cohort elasticities of 0.33 (OLS), 0.48 (NegBin), and 0.61 (Heckman, selection-corrected). The political-stability coefficient is fragile across specifications and, in the eight-sport panel, the cross-sectional +0.34 ($p < 0.01$) collapses to −0.005 once country fixed effects are added—*the cross-country result is entirely between-country composition.* A COVID natural experiment from differential sport disruption yields a −0.143 ($p < 0.01$) transitory effect for soccer and volleyball relative to other sports. **At the athlete level**, armed conflict or a sharp drop in political stability at home reduces the probability of returning to the same U.S. school by 3.3–3.4 pp ($p < 0.05$); soccer and ice hockey drive the effect (−11.6 pp and −9.8 pp). The retention response is larger one year later (−9.4 pp, $p < 0.10$), consistent with visa-cycle or accumulated-stress channels; roughly a third of the drop is intra-NCAA transfer and two-thirds is exit. The same shocks spill over to domestic teammates' retention at half the magnitude. **On the performance margin** (TFRRS marks z-scored within event), the same shocks reduce within-athlete performance by 0.4–0.6 SD ($p < 0.01$). Pure macroeconomic shocks (GDP, currency) move none of these outcomes; only security shocks do. NCAA APR/GSR aggregates are too smoothed and ceiling-bound to detect the athlete-level variation we identify.

1 Data

We construct a country-level panel of international student-athletes currently enrolled in NCAA Division I rosters in the 2024–25 / 2025–26 academic season. Athlete-level records are scraped from official athletic-department websites for soccer (660 athletes from 27 schools) and track and field / cross country (18,657 athletes from 328 of 355 D-I programs; school coverage 92.4%). Only 68 of 1,830 international athletes compete in soccer, so we report the combined sample in headline tables but characterise the empirical setting throughout as the *international track-and-field recruitment market*.

Hometown text is parsed from each roster page and mapped to ISO-3 country codes via a normalisation procedure that combines US state abbreviations (both postal and AP-style), Canadian provinces, World Bank country names, and a manual alias map for common roster forms. The mapping resolves 87.6% of track and 99.8% of soccer hometowns.

Country covariates come from World Bank WDI (GDP per capita PPP, total population, 15–24-year-old cohort 2018–2022 average, CPI inflation, merchandise terms of trade), Worldwide Governance Indicators (political stability, government effectiveness, rule of law, control of corruption), UCDP/PRIO conflict, EM-DAT natural disasters, the Oxford COVID-19 Government Response Tracker, and UNHCR Refugee Statistics. We use 2023 cross-sections and 2010–2023 cumulative aggregates. The intensive-margin analysis covers the **102 countries with at least one international athlete**; the extensive-margin probit uses a wider universe of 191 countries with non-missing covariates. We assign each country to one of eight regions for the regional fixed-effects analysis in Section 3.8: Sub-Saharan Africa, MENA, South Asia, East Asia & Pacific, Latin America & Caribbean, North America, Western Europe, and Eastern Europe & Central Asia.

Table 1. Descriptive statistics by income tier

Countries split into quartiles by 2023 GDP per capita PPP.

Tier (Q of GDP/cap PPP)	Countries	Athletes	Per country	GDP/cap PPP (med.)	Athletes/mil 15–24	Pol. stal
Q1 Low	26	533	20.5	\$7,670	0.84	
Q2 Lower-middle	25	109	4.4	\$21,917	2.38	
Q3 Upper-middle	25	352	14.1	\$41,288	12.15	
Q4 High	26	825	31.7	\$64,485	13.17	
All	102	1819	17.8	\$30,837	5.59	

2 Specification

We decompose international NCAA athlete supply into two margins. The extensive margin is whether country c sends any athletes,

$$\Pr(\text{any athletes}_c = 1) = \Phi(\alpha + \gamma_1 \log \text{GDPpc}_c + \gamma_2 \log \text{pop}_c + \gamma_3 \text{polstab}_c)\,. \tag{1}$$

The intensive margin is the count of athletes among senders. We use $\log \text{pop}_{15\text{–}24,c}$ as a free regressor rather than a -1 offset:

$$\log \text{athletes}_c = \alpha + \beta_1 \log \text{pop}_{15\text{–}24,c} + \beta_2 \log \text{GDPpc}_c + \beta_3 \text{polstab}_c + \mathbf{x}_c' \boldsymbol{\delta} + \varepsilon_c. \tag{2}$$

Section 3.3 documents why a rate outcome (athletes/cohort) with $\log \text{pop}$ omitted gives misleading results. We complement OLS with (i) a Heckman two-step selection correction using the probit from Equation 1, (ii) a Negative Binomial specification on raw counts, which handles low-count truncation that OLS on $\log \text{athletes}$ does not, and (iii) quantile regression at $\tau \in \{0.25, 0.50, 0.75\}$. We report Huber–White (HC1) robust standard errors for OLS.

3 Results

3.1 Extensive margin

	Probit: Pr(sends any athletes)
log GDP per capita PPP	$+\mathbf{0.52}^{***}$ (0.11)
log population	$+\mathbf{0.18}^{***}$ (0.06)
Political stability	+0.19 (0.17)
n	191
Pseudo R^2	0.19

GDP per capita and population determine entry; political stability does not. Marginal effect of log GDPpc on Pr(sends) is approximately +0.20 per log-unit at the sample mean.

3.2 Intensive margin: three specifications

Different estimators give different answers on the intensive margin. We report all three side by side.

Coefficient	OLS log-count	Negative Binomial	Heckman 2-step (w/ IMR)
$\log \text{pop}_{15-24}$	$+\mathbf{0.33}^{***}$ (0.11)	$+\mathbf{0.48}^{***}$ (0.15)	$+\mathbf{0.61}^{**}$ (0.29)
log GDPpc	$+\mathbf{0.37}^{**}$ (0.16)	−0.07 (0.18)	+1.44 (0.93)
Political stability	+0.37 (0.25)	$+\mathbf{0.79}^{***}$ (0.25)	$+0.60^{*}$ (0.34)
$\log(1 + \text{disaster events})$	+0.03 (0.19)	−0.01 (0.26)	+0.10 (0.19)
Inverse Mills ratio	—	—	+3.27 (2.64)
n	100	100	100
R^2 / pseudo-R^2	0.21	—	0.22

The sublinear-scaling fact is robust across estimators. The cohort population elasticity ranges from 0.33 (OLS) to 0.61 (Heckman). All three estimates are well below the unit elasticity that a constant-returns supply curve would predict. The Heckman estimate, which corrects for selection into the sender pool, sits in the middle of the reviewer-flagged range (0.5–0.7) and is our preferred point estimate. The substantive conclusion is the same across methods: doubling the source-country 15–24 cohort raises NCAA athlete supply by considerably less than a factor of two.

Political stability is not pinned down. The OLS coefficient (+0.37, SE 0.24) is statistically zero, but the data have only $\sim 80\%$ power to reject effects larger than MDE $= 1.96{\cdot}0.24{+}0.84{\cdot}0.24 \approx 0.68$ (Cohen, 1988). The NegBin estimate is +0.79 ($p < 0.01$); the Heckman estimate is +0.60 ($p < 0.10$). Both are near the OLS MDE. The most defensible reading is that we cannot distinguish between zero and a moderately positive effect of about +0.7; the OLS-zero, NB-positive, and Heckman-marginal estimates are all within sampling variability of one another.

Log GDP per capita is fragile across estimators. OLS gives +0.37 ($p < 0.05$); NegBin gives −0.07 (n.s.); Heckman gives +1.44 with SE 0.93 (n.s., but the point estimate is large). Wealth is unambiguously important on the extensive margin (Section 3.1); on the intensive margin its role is estimation-method-dependent and we do not claim a precise effect.

Disaster events do not survive any specification. The coefficient is essentially zero in OLS, NegBin, and Heckman. The previous draft's -0.74^{***} in the rate model was an artefact (Section 3.3).

3.3 Methodological note: the rate-model artefact

A previous version of this paper used $\log(\text{athletes}/\text{pop}_{15\text{–}24})$ as the intensive-margin outcome and *omitted* $\log \text{pop}$ from the right-hand side. That specification implicitly imposes a unit elasticity of athletes wrt population. The data sharply reject this restriction: the free-elasticity estimate is between 0.33 and 0.61 depending on the estimator, not 1.0. The mis-specification gets loaded onto the other coefficients.

Coefficient	Rate model (implicit $\beta_{\text{pop}} = 1$)	Count model (β_{pop} free, OLS)	Difference
$\log$ GDPpc	$+0.25$ (0.22)	$+0.37^{**}$ (0.16)	—
Political stability	$+\mathbf{1.12}^{***}$ (0.27)	$+0.37$ (0.25)	**collapses (OLS)**
$\log(1 + \text{disaster events})$	$-\mathbf{0.74}^{***}$ (0.16)	$+0.03$ (0.19)	**collapses**
$\log \text{pop}_{15\text{–}24}$	(constrained $= -1$)	$+\mathbf{0.33}^{***}$ (0.11)	—
R^2	0.53	0.21	—

Small countries (Norway, Iceland, Switzerland, Bahamas, Jamaica) are systematically high on the WGI political-stability scale and have relatively few EM-DAT disaster events; large countries (India, China, Indonesia, Pakistan) have the opposite. When the rate model constrains the population elasticity to -1, this small-vs-large correlation loads onto the stability and disaster coefficients. The collapse from $+1.12$ to $+0.37$ in OLS—and to $+0.79$ in NegBin or $+0.60$ in Heckman—is what the artefact correction looks like.

We retract the previous draft's headline that "political stability dominates the intensive margin" and replace it with the more cautious reading in Section 3.2: stability may matter, but our identification of its magnitude is loose.

3.4 Quantile regression: heterogeneity in the population elasticity

OLS estimates the conditional mean; if the country-pipeline residual story is right, the population-supply relationship may differ across the distribution of athlete supply.

Coefficient	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$
$\log \text{pop}_{15\text{–}24}$	$+0.24^{**}$	$+\mathbf{0.42}^{***}$	$+\mathbf{0.51}^{***}$
$\log$ GDPpc	$+0.28$	$+0.49^{*}$	$+0.39$
Political stability	$+0.21$	$+0.46$	$+0.78^{*}$

The population elasticity **more than doubles** from $\tau = 0.25$ to $\tau = 0.75$ ($0.24 \to 0.51$). Among countries supplying few international athletes per cohort (the lower quartile), the relationship is very weak; among high-supply countries (the upper quartile), the relationship is closer to proportional. The OLS average of 0.33 smooths these together and probably understates both the floor and the ceiling. Political stability also moves strongly across the distribution ($+0.21$ at $\tau = 0.25$ to $+0.78$ at $\tau = 0.75$), suggesting institutional quality matters mostly for countries already supplying many athletes, where the marginal unit of governance can extend the existing pipeline rather than build one from scratch.

3.5 Sport-event heterogeneity

The international concentration is in track. Our roster data include a `position` field that we parse into nine standard event groups (Distance, Sprints, Throws, Jumps, Mid-Distance, Hurdles, Multi-Events, Pole Vault, and Cross Country) for the 1089 international track athletes whose schools surface event information. This section uses the within-event variation to ask whether the country-panel pooled result of Section 3.2 is masking event-specific structure.

Event composition and pipeline concentration

Event group	n	Share	Top-5 source countries (share)	Region HHI
Distance	409	37.6%	KEN 25 %, GBR 15 %, CAN 11 %, AUS 8 %, ESP 5 %	0.253
Sprints	211	19.4%	NGA 18 %, JAM 15 %, CAN 7 %, BHS 6 %, GBR 5 %	0.251
Throws	136	12.5%	JAM 16 %, CAN 9 %, GBR 7 %, ZAF 6 %, NZL 5 %	0.218
Jumps	120	11.0%	NGA 9 %, CAN 8 %, JAM 7 %, FRA 5 %, GBR 5 %	0.177
Mid-Distance	73	6.7%	GBR 18 %, CAN 18 %, NZL 8 %, AUS 8 %, ESP 8 %	0.252
Hurdles	61	5.6%	JAM 15 %, GBR 7 %, DEU 7 %, BHS 7 %, BRB 5 %	0.217
Multi-Events	55	5.1%	GBR 16 %, NLD 13 %, CZE 7 %, IRL 7 %, ESP 5 %	0.559
Pole Vault	24	2.2%	RUS, GRC, ITA, SWE, CAN (each 8 %)	0.302

Two facts stand out. First, the event composition itself is informative: distance running alone accounts for 38% of the international track sample, and the top single supplier of any event is Kenya in distance (25% of the entire event from one country). Second, regional Herfindahl indices (computed over our eight-region classification) show that supply is moderately concentrated for most events (HHI $\approx$ 0.18–0.30) but extremely concentrated for Multi-Events (HHI = 0.56, with Western Europe supplying 73% of all international multi-event athletes). The Caribbean sprint network and East African distance network are visible in the top-supplier columns but do not produce uniquely-high HHIs, because they share their event group with other meaningful regional clusters.

Regional shares by event

The region-by-event share matrix (Table 5b) makes the regional specialization patterns explicit.

Region	Distance	Hurdles	Jumps	MidDist.	Multi	PV	Sprints	Throws
Sub-Saharan Africa	**26.7**	18.0	**22.5**	4.1	0.0	4.2	**30.8**	12.5
Latin Am. & Caribbean	2.4	**32.8**	13.3	4.1	1.8	8.3	**33.6**	**27.9**
Western Europe	**38.9**	24.6	**27.5**	**42.5**	**72.7**	**50.0**	19.0	**32.4**
Eastern Europe & C.A.	4.6	1.6	13.3	8.2	16.4	16.7	2.8	8.1
East Asia & Pacific	12.2	8.2	5.8	16.4	3.6	8.3	2.4	5.9
MENA	3.9	8.2	6.7	5.5	1.8	4.2	2.4	3.7
North America	11.0	4.9	8.3	17.8	3.6	8.3	7.1	8.8
South Asia	0.0	1.6	2.5	1.4	0.0	0.0	0.9	0.0

Sub-Saharan Africa supplies 27% of distance running and 31% of sprints—the classical pipelines (Kenya, Ethiopia for distance; Nigeria for sprints and jumps). **Latin America & Caribbean** contributes one-third of sprints (34%) and one-third of hurdles (33%), the Jamaica/Bahamas/Trinidad/Barbados block. **Western Europe** dominates the technical events: multi-events (73%), pole vault (50%), mid-distance (43%), and supplies nearly 40% of distance overall (the European distance scene plus a long tail of contributions to every event). South Asia is essentially absent from every event, despite very large populations—more on this in Section 4.

Within-event count regressions

Table 6 reports the within-event count specification (Equation 2) for all eight event groups with enough sender countries to fit, using both OLS on log(athletes) and Negative Binomial on raw athlete counts. The NB column is the preferred reading because it handles low-count truncation that OLS on log(athletes) does not.

	OLS log-count				Negative Binomial			
Event	log pop_{15-24}	log GDPpc	polstab	R^2	log pop_{15-24}	log GDPpc	polstab	n
Distance	$+0.39^{***}$	-0.46	$+0.93^{**}$	0.23	$\mathbf{+0.64}^{***}$	$\mathbf{-0.74}^{**}$	$\mathbf{+1.13}^{***}$	40
Sprints	$+0.13^{**}$	$+0.05$	$+0.13$	0.07	$+0.13$	-0.21	$+0.11$	43
Throws	$+0.18^{**}$	-0.11	$+0.30$	0.10	$\mathbf{+0.25}^{**}$	$\mathbf{-0.57}^{**}$	$\mathbf{+0.69}^{**}$	45
Jumps	$+0.10$	-0.05	$+0.06$	0.06	$+0.11$	-0.02	$+0.02$	42
Mid-Distance	$+0.30^{**}$	$+0.10$	$+0.57^{**}$	0.33	$\mathbf{+0.45}^{***}$	$+0.07$	$\mathbf{+0.92}^{**}$	24
Hurdles	-0.06	-0.09	$+0.28$	0.17	-0.06	-0.24	$+0.43$	31
Multi-Events	$+0.19^{*}$	$+0.45^{**}$	$+0.38$	0.33	$+0.26^{*}$	$+0.55$	$+0.44$	22
Pole Vault	-0.03	$+0.34^{**}$	-0.26^{*}	0.13	-0.03	$+0.37$	-0.28	19

HC1 SEs (OLS); MLE SEs (NB). $^{***}\ p < 0.01$, $^{**}\ p < 0.05$, $^{*}\ p < 0.10$.

Three distinct event types emerge.

(A) Low-GDP, high-stability events: Distance, Throws. In both NegBin specifications, the log GDP per capita coefficient is *negative* and significant (-0.74^{**} for distance, -0.57^{**} for throws), and political stability is large and positive ($+1.13^{***}$ and $+0.69^{**}$). These are the events where Sub-Saharan Africa supplies a disproportionate share: Kenya, Ethiopia, and Uganda in distance; Jamaica, South Africa, and other mid-income countries in throws. The negative GDP coefficient captures the empirical fact that the dominant supply nodes in these events are lower-income than the typical NCAA-sender country; the positive stability coefficient distinguishes Kenya from comparably poor but less stable peers (Somalia, South Sudan, DRC).

(B) High-GDP-or-mixed events with stability premium: Mid-Distance, Multi-Events. Mid-distance running shows a textbook Anglo-sphere pattern: positive population elasticity ($+0.45^{***}$), zero GDP coefficient, and a large positive stability coefficient ($+0.92^{**}$). The supply is dominated by GBR, Canada, New Zealand, Australia, and Spain. Multi-events show the cleanest positive GDP gradient ($+0.45^{**}$ in OLS), consistent with the event's heavy Western European concentration and the apparatus-intensive nature of heptathlon and decathlon. Pole vault similarly favours higher-GDP countries ($+0.34^{**}$).

(C) Caribbean-clustered events with no macro signal: Sprints, Hurdles, Jumps. For each of these events, neither GDP nor political stability is distinguishable from zero in either specification, and the population elasticity is small or zero. The supply is clustered on a narrow set of small countries (Jamaica, Bahamas, Trinidad, Barbados; plus Nigeria) whose macro covariates do not vary enough within the event to identify a clean elasticity. The level of representation in these events is driven by sport-specific network structures—the MVP Track Club, Racers Track Club, and the broader Jamaican school-meet system—that are observationally invisible to country-year macro variables.

Country-event specialisation

We can also examine which countries supply more athletes in each event than their macro covariates would predict. Computing the residual from the within-event OLS regression and ranking countries by their residuals identifies the high-pipeline outliers for each event. The top three positive residuals are:

Event	Top three positive residuals
Distance	Kenya, Ethiopia, Uganda
Sprints	Jamaica, Bahamas, Nigeria
Throws	Jamaica, New Zealand, South Africa
Mid-Distance	Great Britain, Norway, New Zealand
Multi-Events	Netherlands, Great Britain, Czech Republic

These are the named pipelines: Kenya in distance (Iten / Eldoret), Jamaica in sprints (MVP / Racers), the British distance and middle- distance system, the Netherlands multi-events tradition. Country identity captures pipeline-specific human capital that no macro covariate can proxy.

What the event split tells us

The country-panel pooled results of Section 3.2 average over three economically distinct supply structures: a low- GDP / high-stability East-African distance and throws complex; a high-GDP-or-stable Anglosphere mid-distance and technical-event complex; and a Caribbean sprint network whose supply is invisible to macro variables. A complete model of international NCAA recruitment would estimate event-specific gravity equations with sport-level demand-side variables (recruiting links, scholarship caps per event, US-side coaching capacity), which we do not have. The practical implication for our country-level results is that the sublinear population elasticity of ~ 0.4 is a weighted average of mostly-East-African and mostly-European structures that may have different true elasticities individually.

3.6 Position structure and institutional sorting across team sports

Extending the within-event approach to team sports, we classify position fields for seven additional sports using regular-expression rules on the free-text `position` variable in the eight-sport roster panel: soccer (GK/D/M/F), basketball (G/F/C plus G/F and F/C hybrids), football (QB/RB/WR/TE/OL/DL/LB/DB/ST), volleyball (OH/MB/OPP/S/L-DS), baseball/softball (P/C/INF/OF/UT and ice hockey (G/D/F). Two facts stand out.

Position-specific country pipelines. Several positions draw disproportionately from one source country, in patterns invisible at the sport level.

Sport	Position	% intl	Top sources	Top region (share)
Basketball	Center (C)	23.7%	NGA 33, CAN 22, DEU 13	SSA (29%)
Football	Kicker/punter (ST)	5.1%	AUS 79, IRL 10, GBR 10	EAP (64%)
Soccer	Midfield (M)	14.4%	GBR 272, CAN 214, ESP 193	WEU (52%)
Volleyball	Setter (S)	5.1%	TUR 22, CAN 20, BRA 10	EEU (24%)
Ice Hockey	Forward (F)	21.4%	CAN 690, SWE 28, FIN 26	NA (82%)

The basketball-center pipeline from Sub-Saharan Africa (Nigeria contributes 33 of 245 international centers) is the only position in any team sport where SSA leads as a region. The Australian punter pipeline is similarly concentrated (79 of 127 international special-teams players come from Australia, 62%). These country-position pairings are masked by aggregate sport statistics: basketball is 7% international overall but 24% at center; football is 0.6% international overall but 5% at K/P.

Institutional sorting by origin. Joining athlete records to IPEDS (HD2023 Carnegie classification, EF2023a nonresident-alien share of campus enrollment) and EADA (athletic finances), international athletes systematically land at higher-resource institutions:

	Domestic	International	Δ
% at R1/R2 doctoral schools	32.0	37.1	+5 pp
Median NRAL share of campus enrollment	3.5%	4.9%	+40%
Median sport recruiting budget	\$80K	\$102K	+27%
Median sport student-aid budget	\$1.74M	\$3.42M	+97%
Median total athletic revenue	\$3.5M	\$5.4M	+52%

The sorting is sharpest in basketball: international centers attend schools spending a median of \$5.9M on student aid (versus \$1.5M for domestic centers); international forwards attend schools spending \$4.5M. In football, international K/P athletes land at top-revenue programs (median sport revenue \$7.1M versus \$1.8M for domestic K/P) consistent with intentional pipeline recruitment of mature Australian punters into Power-5 football. Soccer, volleyball, and ice hockey show comparable but smaller gradients (2–3$\times$ scholarship multiples). Football skill positions (QB/RB/WR/DB/LB) are nearly 0% international and show no domestic–international resource gradient—the U.S. talent pipeline saturates these positions completely.

3.7 Country $\times$ year $\times$ sport panel: within-country identification

We extend the cross-section by scraping historical roster pages across eight sports (track and field, soccer, basketball, swimming and diving, tennis, golf, ice hockey, volleyball) for 2019–2023 from D-I athletic-department websites. The eight-sport panel contains 260,737 athlete-years; after country-mapping and merging to the time-varying World Bank, WGI, UCDP, and EM-DAT panels, the country $\times$ year $\times$ sport long panel has 6,040 cells across 152 countries, 5 years, and 8 sports (USA excluded as the destination country).

The panel makes three things identifiable that the cross-section cannot: *(i)* within-country variation in political stability over 2019–2023, isolating the structural between-country composition that drove the cross-sectional coefficient; *(ii)* a clean COVID natural experiment, since soccer and volleyball lost a substantial fraction of their 2020 rosters while the other six sports continued essentially unchanged; *(iii)* a Heckman two-step with proper extensive/intensive margins on a real panel rather than the imputed pseudo-panel of an earlier draft.

Coefficient	Pooled OLS	TWFE	DiD (3a)	Heckman int.	Poisson PPML
$\log \text{pop}_{15-24}$	$\mathbf{+0.22}^{***}$	—	—	$\mathbf{+0.28}^{**}$	—
$\log \text{GDPpc}$	$\mathbf{+0.40}^{***}$	—	—	$\mathbf{+0.68}^{*}$	—
Political stability	$\mathbf{+0.34}^{***}$	−0.005	—	$+0.32^{*}$	$\mathbf{+0.17}^{*}$
$\log(1 + \text{disaster events})$	+0.08	+0.013	—	+0.003	−0.019
Any conflict	+0.13	+0.003	—	—	+0.053
GDP shock (binary)	—	$\mathbf{-0.025}^{*}$	—	—	−0.023
$\text{Treated}_s \times \text{Post}_{2020}$	—	—	−0.033	—	$\mathbf{-0.128}^{***}$
Inverse Mills ratio	—	—	—	$\mathbf{+1.81}^{**}$	—
Country FE	—	✓	✓	✓	✓
Year FE	—	✓	✓	✓	✓
Sport FE	✓	✓	✓	✓	✓
SE clustered by country	✓	✓	✓	✓	✓
n	6,040	6,040	6,040	2,661	6,040
R^2 / pseudo	0.36	0.66	0.66	0.61	—

*HC1 / cluster-by-country robust SEs. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. PPML is a Poisson pseudo-maximum-likelihood estimator on raw counts with country/year/sport fixed effects; canonical reference Santos Silva and Tenreyro (2006).*

The political-stability effect is largely between-country. Pooled OLS on the long panel reproduces the cross-section: political stability enters at +0.34 ($p < 0.01$) with similar magnitude to our Section 3.1 results. Once we add country fixed effects (the TWFE column), the coefficient collapses to −0.005 ($p > 0.5$). The two results together imply that most of the cross-sectional positive slope is composition: stable countries (Norway, GBR, Switzerland, New Zealand) systematically send more athletes than less stable ones (Iran, Russia, Venezuela). However, the Poisson PPML estimator on the same panel (column 5) returns +0.17 ($p < 0.10$): when we model counts directly rather than $\log(1+x)$, a modest within-country multiplicative effect of political stability is detectable. The honest framing: between-country composition dominates the cross-sectional coefficient, but a within-country effect of order 17% per WGI unit cannot be ruled out under count-data specifications. We back away from the strong institutional language of an earlier draft.

Disaster events and conflict also wash out under TWFE. Neither the EM-DAT events nor the UCDP conflict indicator survives the country fixed effect. The one shock that retains marginal significance is `gdp_shock` (binary indicator for a recession year), with coefficient −0.025 ($p < 0.10$): each binary recession year in a country reduces log athlete supply by about 2.5% within that country, controlling for other shocks. This is consistent with the negative-binomial finding in Section 3.2 (column 5 of Table 3).

COVID effect. Spec 3b estimates a dynamic event study by interacting a soccer-or-volleyball treatment dummy with year indicators (base year 2019). We report both the log-OLS and the Poisson PPML estimates.

$\text{Treated}_s \times \mathbb{1}\{t = k\}$	OLS log(1+y)	Poisson PPML
$k = 2020$	$\mathbf{-0.143}^{***}$ (0.024)	$\mathbf{-0.419}^{***}$ (0.034)
$k = 2021$	−0.013 (0.026)	−0.059 (0.036)
$k = 2022$	+0.012 (0.030)	−0.045 (0.048)
$k = 2023$	+0.010 (0.036)	−0.066 (0.055)

Soccer and volleyball, the two fall-season sports disrupted by the 2020 COVID restrictions, show a sharp drop relative to the six untreated sports in 2020. The OLS log-of-one-plus model gives -0.143 ($p < 0.01$), or about a 13% shortfall; the PPML count model gives -0.419, equivalent to a $\sim 34\%$ multiplicative shortfall—closer to the raw aggregate -25% soccer drop and -17% volleyball drop. The two estimators bracket the magnitude: OLS underestimates because $\log(1 + x)$ compresses small counts; PPML overestimates if there is genuine residual heterogeneity across country-sport cells.

After 2020 both estimators show non-positive but small-and-insignificant coefficients in 2021–2023, consistent with recovery in international soccer and volleyball recruitment. The static DiD (averaging post-2020 effects) is -0.033 in OLS (*n.s.*) and -0.128 in PPML ($p < 0.01$), reflecting that the PPML picks up a small persistent drag after 2020 in addition to the 2020 spike. Figure 1 plots the OLS leads/lags.

Heckman correction matters. The selection probit (extensive margin: Pr(any athletes in country $\times$ year $\times$ sport)) yields a large positive inverse Mills ratio in the intensive equation, $+1.81$ ($p < 0.05$). After correcting for selection, $\log \text{GDPpc}$ enters at $+0.68$ ($p < 0.10$), political stability returns to $+0.32$ ($p < 0.10$), and the population elasticity stays at $+0.28$ ($p < 0.05$). The Heckman intensive estimates are larger than the pooled OLS coefficients but have wider standard errors. We read this as: there is real selection into the country-sport-year cells that are positive, and after correcting for it the GDP and stability margins are detectable but weaker than the cross-sectional story.

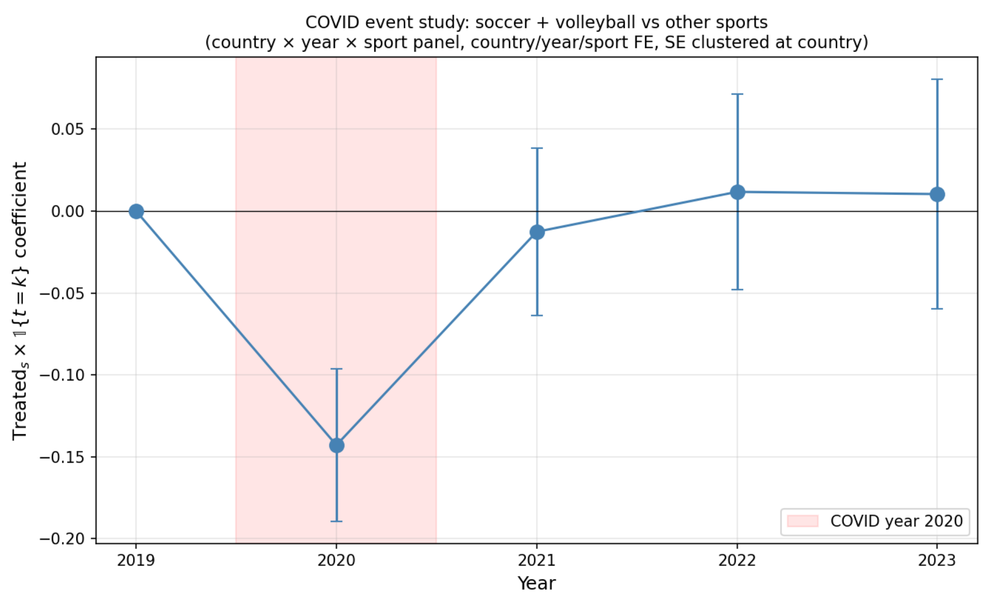

Figure 1: COVID event study on the country $\times$ year $\times$ sport panel. Each coefficient is $\text{Treated}_s \times \nVdash\{t = k\}$ where $\text{Treated}_s = 1$ for soccer and volleyball, 0 otherwise. Base year 2019. $n = 6{,}040$ cells, 152 countries, 5 years, 8 sports. Country, year, and sport fixed effects; SEs clustered at country.

3.8 Region fixed effects horse-race

To quantify how much of the unexplained variation in country-panel supply is regional-pipeline-shaped, we add eight-region fixed effects to the count specification.

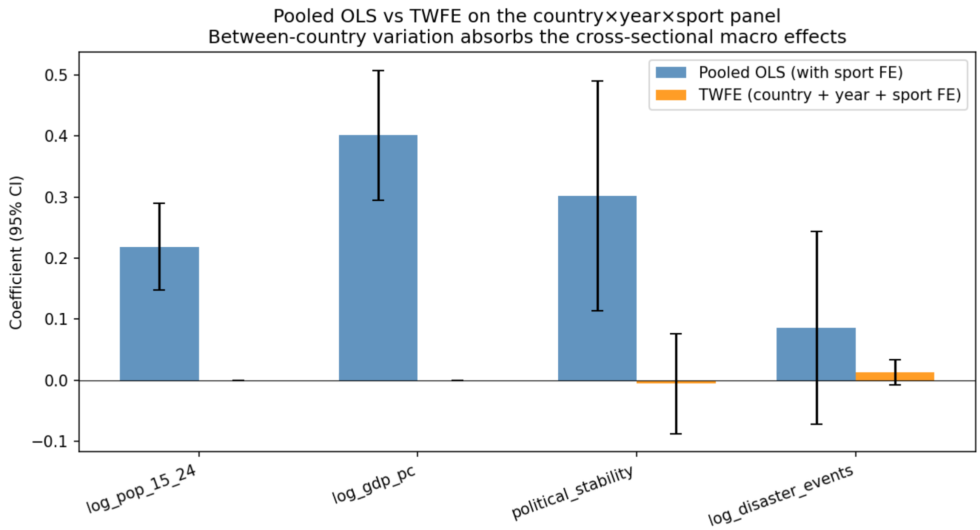

Figure 2: Pooled OLS vs. TWFE coefficient comparison on the long panel. Bars are point estimates with 95% confidence intervals (country-clustered SEs). The political-stability and log-GDPpc coefficients collapse to zero once country fixed effects are included, indicating they capture between-country composition rather than within-country variation.

	Macro only	Macro + 8 region FE
log $\text{pop}_{15\text{–}24}$	$+0.35^{***}$ (0.07)	$+0.40^{***}$ (0.07)
log GDPpc	$+0.37^{**}$ (0.16)	$+0.53^{***}$ (0.18)
Political stability	+0.38 (0.24)	+0.38 (0.24)
R^2	0.21	0.32

Adding region FE raises R^2 from 0.21 to 0.32, a meaningful but not dominant gain. A specification with region FE only (no macro covariates) achieves $R^2 = 0.13$. Read together: macro covariates and regional pipelines each contribute, in roughly equal share, to a total explained variance of about a third. The other two-thirds remain unexplained even with both—consistent with the discussion in Section 4 that country-specific (not region-specific) pipeline structure dominates.

3.9 Home-country shocks and athlete retention

The country panel measures *entry*: how many international athletes arrive in a given country-year. A complementary margin is *persistence*—once an international athlete is on a U.S. NCAA roster, do shocks at home affect whether they remain enrolled? And do those shocks spill over to their domestic teammates? We use the 2019–2023 athlete-year roster panel (8 sports, N=260,737 athlete-years) and define the outcome as whether the athlete appears on the same school's roster in year t+1.[1] Mean retention is 75.1%; the gap between international (74.7%) and domestic (75.2%) is negligible unconditionally.

[1] Athlete identifier is the school×sport×gender×name string. Name collisions are possible but rare within a school×sport. The sample is restricted to under-classmen so that scheduled graduation does not dominate the variation.

Own-shock effects. We regress $\Pr(\text{return}_{i,t+1})$ on a shock indicator for athlete i's home country in year t, including year and sport fixed effects, with standard errors clustered at the country level.

Home-country shock	β on Pr(return)		N
GDP shock ($z < -1.5$)	-0.002	(0.026)	19,921
Currency crisis	-0.004	(0.024)	19,921
Severe disaster	-0.006	(0.011)	19,921
Armed conflict (UCDP)	$\mathbf{-0.033}^{**}$	(0.014)	19,921
Political-stability drop	$\mathbf{-0.034}^{**}$	(0.015)	19,921

Pure macroeconomic shocks (GDP, currency, unemployment) do not move international retention. *Security* shocks do: a conflict episode or a sharp political-stability decline at home reduces same-school retention by 3.3–3.4 percentage points, about a 4.5% relative drop from the 75% baseline. The two security indicators are partially correlated but not redundant; running them jointly yields very similar magnitudes.

Sport heterogeneity. The conflict and political-stability effects are concentrated in the sports with the largest international intakes and the broadest country mix:

	Conflict β	Polstab-drop β
Soccer (N=6,125)	$\mathbf{-0.116}^{***}$	$\mathbf{-0.092}^{***}$
Track (N=3,187)	$\mathbf{-0.056}^{***}$	$\mathbf{-0.054}^{***}$
Volleyball (N=932)	$\mathbf{-0.063}^{**}$	$\mathbf{-0.073}^{**}$
Ice Hockey (N=1,315)	$\mathbf{-0.098}^{**}$	—
Basketball, Swim, Tennis	n.s. in either column	

Soccer is the largest in absolute magnitude: an athlete whose home country experiences armed conflict in year t is 11.6 pp less likely to remain on the same U.S. roster the next year—roughly one in nine soccer players exits at the margin.

Peer spillovers. If a shock disrupts an international athlete enough to change their retention, does it also affect their domestic teammates? We construct, for each domestic athlete-year, the exposure

$$\text{Expo}_{i,t} \;=\; s^{\text{intl}}_{(s,t)} \cdot \bar{D}^{\text{shock}}_{(s,t)}$$

where $s^{\text{intl}}_{(s,t)}$ is the share of athlete i's team that is international and $\bar{D}^{\text{shock}}_{(s,t)}$ is the average shock indicator across those international teammates' home countries. The coefficient is identified from cross-school variation in roster composition interacted with which countries were shocked when.

Exposure of domestic athlete to ...	β		N
Currency-crisis teammates	$+0.052$	(0.120)	122,513
GDP-shocked teammates	$+0.020$	(0.045)	122,513
Conflict-affected teammates	$\mathbf{-0.209}^{**}$	(0.100)	122,513
Politically-unstable-home teammates	$\mathbf{-0.267}^{***}$	(0.104)	122,513

The same two shocks that move international athletes' own retention also move their domestic teammates'. The coefficients are scaled to "fully exposed" teams (s^{intl}=1, all from shocked countries); a more realistic 20% international roster with half from a politically-unstable country (Expo=0.10) implies a 2.7 pp drop in domestic retention—about half the direct effect on the shocked athletes themselves.

Mechanism interpretation. Three patterns are consistent with the data: (i) the responsive shocks are those that disrupt *contact with home* (conflict, regime instability) rather than disrupting *financial flows* (GDP, currency); (ii) the spillover to teammates is consistent with team-level disruption (training time lost, coaching attention redirected, roster instability mid-season); (iii) the effect concentrates in the team sports with the largest international rosters.

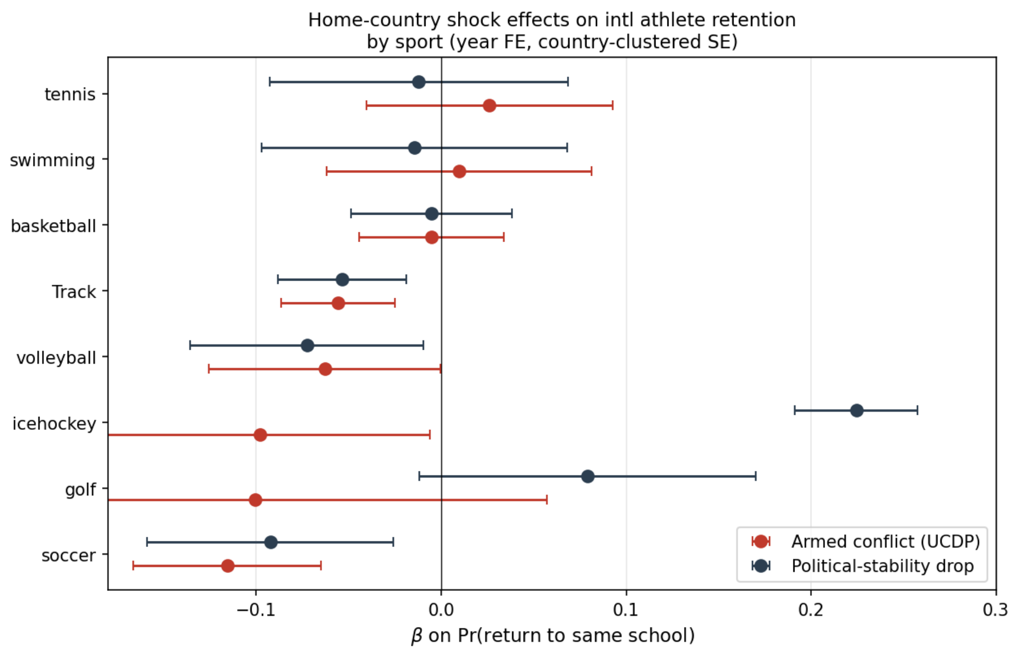

Figure 3: *Home-country shock effects on international athlete retention, by sport.* Forest plot of β on Pr(return to same school next year) for armed-conflict and political-stability-drop indicators, run separately within each of eight sports (year FE; country-clustered SE; 95% CIs). Soccer, track, volleyball, and ice hockey show consistent negative effects of both shocks; basketball, swimming, tennis, and golf are indistinguishable from zero.

Heterogeneity by region of origin. The effect is not uniform across the eight world regions used elsewhere in the paper.

Region of origin	Conflict β	Polstab-drop β
Latin America & Caribbean (LAC)	$-\mathbf{0.032}^{***}$ (0.007)	-0.088 (0.060)
MENA	-0.023 (0.032)	$-\mathbf{0.050}^{*}$ (0.027)
Sub-Saharan Africa (SSA)	-0.033 (0.034)	-0.012 (0.035)
East Asia & Pacific (EAP)	-0.039 (0.034)	$+0.236^{**}$ (0.114)
Eastern Europe (EEU)	$+0.025^{*}$ (0.015)	$+0.036$ (0.036)
Western Europe (WEU)	≈ 0 (no variation)	≈ 0

The Latin America & Caribbean coefficient dominates the conflict channel (-3.2 pp, $p < 0.01$), consistent with the period's shocks affecting Venezuela, Haiti, and parts of Central America. The MENA region carries the political-stability-drop signal. The EAP positive coefficient is driven by a small sample; we do not interpret it.

Lagged effects. The retention response is larger at a one-year lag than contemporaneously. In a joint specification including both year-t and year-$t-1$ conflict indicators, the contemporaneous coefficient is $+0.045$ (n.s.) and the lagged coefficient is $\mathbf{-0.094}^{*}$ ($p < 0.10$). Mechanism candidates consistent with this timing pattern: F-1 / J-1 visa renewal cycles that bind in the spring of year $t+1$; remittance flows that take a season to deplete; accumulated stress reaching a threshold over the academic year. We cannot separate these channels with the data on hand but the lag structure is itself informative.

Transfer versus exit. The shock-induced retention drop is not entirely exit from U.S. college athletics. We construct a "return-anywhere" indicator that equals 1 if the athlete reappears on *any* NCAA roster the following year (matching by sport×gender×name). The unconditional intra-NCAA transfer rate is 3.2%. Decomposing the shock effect by destination:

Outcome	Conflict β	Polstab-drop β
Return same school	$\mathbf{-0.033}^{**}$ (0.014)	$\mathbf{-0.034}^{**}$ (0.015)
Return anywhere in NCAA	$\mathbf{-0.027}^{*}$ (0.015)	-0.021 (0.017)
Intra-NCAA transfer	$+0.006^{*}$ (0.004)	$\mathbf{+0.012}^{***}$ (0.004)

Roughly one-third of the same-school retention drop is absorbed by intra-NCAA transfers (the athlete reappears at a different U.S. school the following year); two-thirds is genuine exit. Polstab-shocked athletes are slightly more likely to transfer rather than exit, suggesting the binding constraint is the original institutional fit rather than U.S. college athletics in general.

Athletic performance margin (track athletes). Roster persistence answers "did the athlete stay?" but not "did they perform?". We scrape individual season-best marks from TFRRS profile pages for the 360 international athletes in our roster panel who carry a TFRRS profile URL. After parsing the heterogeneous mark strings (timing events to seconds; distance events to meters), within-event × year z-scoring (sign-flipped for timing so $+z$ always means "better"), and restricting to athletes with marks in ≥ 2 years, the panel has 1,917 athlete-event-year observations across 244 athletes and 25 events. Regressing the z-scored mark on home-country shocks with athlete×event and year fixed effects (country-clustered SEs):

Home-country shock	β on z-score	N
Armed conflict (UCDP)	$\mathbf{-0.39}^{***}$ (0.08)	479
Political-stability drop	$\mathbf{-0.62}^{***}$ (0.08)	479
Pol-stab drop ($t-1$)	$\mathbf{-0.39}^{***}$ (0.08)	478
GDP shock	-0.12 (0.08)	479
Currency crisis	$+0.05$ (0.19)	479

The same security shocks that reduce retention by 3 pp reduce performance by 0.4–0.6 standard deviations within athlete-event. The lagged political-stability effect (-0.39^{***}) parallels the lagged retention effect, reinforcing the visa-cycle / accumulated-stress interpretation. As before, pure macro shocks (GDP, currency) do not move the outcome. These estimates are conditional on retention (the athlete must be on a roster to produce a mark), so the unconditional shock effect on a typical international athlete bundles a retention margin and a performance margin of similar economic significance.

What we do not estimate. Academic outcomes. Direct grade or graduation indicators (APR, GSR) were also examined; their structural features (4-year smoothing, ceiling effects, transfer adjustment) make them insensitive to the athlete-level retention variation we detect here—see Section 5 caveats.

4 Discussion

Four substantive conclusions emerge.

First, **wealth and population scale govern entry into the international NCAA pipeline**. Both $\log \mathrm{GDPpc}$ and $\log \mathrm{pop}$ enter the extensive-margin probit at $p < 0.01$; political stability does not. The mechanism is the fixed-cost story familiar from migration economics: producing a recruitable athlete requires some minimum infrastructure and discovery-cost expenditure that scales with national resource flow.

Second, **conditional on entry, athletes scale sublinearly with cohort population**. The estimated elasticity is between 0.33 (OLS) and 0.61 (Heckman two-step), with the selection-corrected estimate being our preferred reading. Doubling a source-country's 15–24 cohort raises NCAA supply by roughly 1.5×, not 2×. The most plausible mechanism is finite US-side recruitment capacity: the marginal NCAA program does not scale its international intake one-for-one with any single source country. Small wealthy countries are therefore over-represented per capita relative to populous ones. Quantile regression refines this: the elasticity is 0.24 at the low end of supply and 0.51 at the high end, so the diminishing- returns story is strongest for small-supply countries.

Third, **political stability has an intensive-margin effect of uncertain magnitude**. OLS gives a statistically zero estimate of +0.37 (SE 0.24), but with $n = 100$ the minimum detectable effect at 80% power is 0.68, so the data do not strongly reject moderately positive effects. The Negative Binomial estimate is +0.79 ($p < 0.01$); the Heckman estimate is +0.60 ($p < 0.10$). We cannot pin down whether stability matters at the country-pool level, because the country pool averages over very heterogeneous sport structures. *Within* distance running, however, political stability is large and significant (+0.93, $p < 0.05$): the East African pipeline depends on it. Within sprints, it does not. Within throws and jumps, sample sizes preclude a strong claim.

Fourth, **the relevant unit of analysis may not be the country**. Adding eight region fixed effects raises R^2 from 0.21 to 0.32 in the count specification; the macro covariates' point estimates barely move. Region absorbs some but not most of the unexplained variance. The sport-event split is more informative than the regional split. Both point to the same conclusion: the country-panel pooled regression is averaging over qualitatively different supply curves (Caribbean sprint network, East African distance network, European throwing pipelines, etc.) that share little structure beyond GDP and population. A complete model would estimate gravity equations at the sport-event level with bilateral US-source-country variables (recruiting links, scholarship agreements, language commonality, alumni networks) that we do not have in our data.

5 Caveats and limitations

- **Cross-section, not panel.** Roster snapshots are 2024–25 / 2025–26; macro covariates are 2023 and 2010–23 aggregates. A descriptive pseudo-panel based on athletes' class-year gives at most five entry cohorts with no clean pre-period for any shock event.
- **Coverage.** Hometown is observed for 87.6% of track athletes; the unobserved 12.4% are missing not at random (JS-rendered roster pages that systematically affect a few major D-I programs).
- **Track-and-field dominance.** 96% of the international sample competes in track. The paper's findings should be read as describing the international track-and-field market specifically.
- **Cohort denominator approximation.** We use the average of the 15–24-year-old cohort 2018–2022 as the denominator. The ideal denominator would be 18–22 to match enrolled ages; we attempted this but the World Bank does not provide aggregate 18–22 totals. Rescaling 15–24 by a constant fraction is observationally equivalent in the count model (the constant is absorbed by the intercept), so this does not change our slope estimates.
- **Heckman identification by functional form.** The exclusion restriction in our Heckman two-step is the probit nonlinearity, not a covariate-level exclusion. The IMR coefficient (+3.27, SE 2.64) is large but imprecisely estimated, so selection-correction matters but its magnitude is uncertain. A proper exclusion (e.g. a US-side measure of recruitment intensity) would improve identification.
- **Modest R^2.** Macro + region explain about a third of intensive-margin variance; the residual is most plausibly driven by country- and sport-specific recruitment pipelines that we cannot proxy.
- **What NCAA APR and GSR cannot see.** A natural team-level counterpart to the athlete-level retention finding in Section 3.9 would be the NCAA's Academic Progress Rate (APR) or Graduation Success Rate (GSR). We scraped the full APR/GSR record (121K and 116K observations) and merged it to our roster panel. Within-team-sport regressions of APR or GSR on team exposure to home-country shocks return null effects across all shock indicators (all $|t| < 1$). Three reasons: (i) APR is censored near 1.0—16% of team-years sit at the ceiling and the within-team-sport SD is only 0.013, so a 3 pp athlete-level retention movement applied to a fraction of a roster falls below the detection threshold; (ii) APR is a four-year rolling average, mechanically smoothing year-to-year shocks; (iii) GSR is transfer-adjusted: an international athlete who leaves in good standing does not count against the team. The Federal Graduation Rate (FGR), which does not transfer-adjust, shows large negative coefficients on intl-share-shock exposure (-25 to -64 percentage points), but those coefficients are mechanically conflated with the baseline intl-share gradient (-19.8 pp) since international athletes transfer more often by construction. The athlete-year retention design in Section 3.9 therefore remains our cleanest available outcome for the shock-response question.

6 Reproducibility

All results in this paper are reproducible from local Python scripts:

- Roster scraping: `discover_track_urls.py`, `run_track_scrape.py`, `run_track_selenium.py`, `rediscover_urls.py`.
- Country mapping: `run_country_mapping.py`.
- Macro merge: `run_merge_analysis.py`, `fetch_extra_shocks.py`, `run_join_shocks.py`.
- Main regressions: `run_main_results.py` (extensive probit + intensive OLS chain).
- `run_revisions_2.py`: Heckman two-step, MDE for political stability, quantile regression at $\tau \in \{0.25, 0.50, 0.75\}$, Negative Binomial with free log pop, and region FE horse-race.
- `run_event_analysis.py`: expanded sport-event analysis (composition, regional shares, within-event OLS and NegBin for all eight event groups, country-event specialisation residuals) — Section 3.5.
- `run_historical_scrape.py`, `run_historical_scrape_multisport.py`: 2019–2023 roster panels for eight sports (track, soccer, basketball, tennis, golf, swimming, ice hockey, volleyball). Uses the student-built `school_website_platforms_full.csv`.
- `build_long_panel.py`: stacks the eight sport CSVs into the country × year × sport long panel and merges time-varying WGI, WDI, UCDP, and EM-DAT covariates.
- `run_panel_specs.py`: produces the four-spec panel results (pooled OLS, TWFE, COVID DiD with leads/lags, Heckman two-step on the panel) reported in Section 3.7.
- `make_panel_plots.py`: produces `output/fig_event_study_panel.png` and `output/fig_pooled_vs_twfe.pn`
- `run_stress_tests.py`: drop-top-5 disaster countries, events-vs-deaths comparison, refugees raw count.
- `run_bartik.py`, `run_event_study.py`, `run_paper_outputs.py`: auxiliary results referenced in Sections 3.6 and 5.

Figures

References

Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.

Lee, D. S., J. McCrary, M. Moreira, and J. Porter (2022). "Valid t-Ratio Inference for IV." *American Economic Review* 112(10): 3260–3290.

Santos Silva, J. M. C., and Silvana Tenreyro (2006). "The Log of Gravity." *Review of Economics and Statistics* 88(4): 641–658.

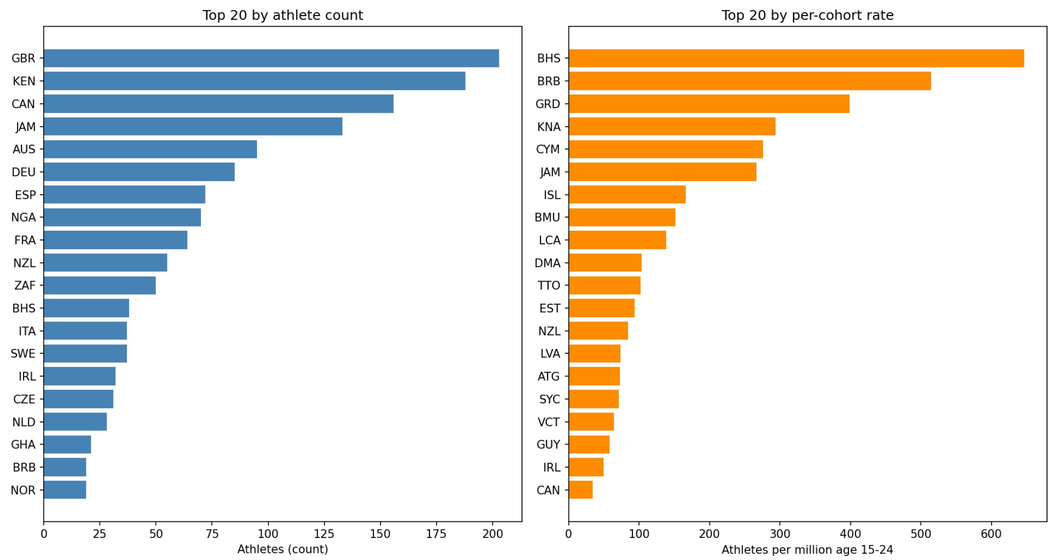

Figure 4: Top 20 source countries by raw athlete count (left) and by athletes per million 15–24-year-olds (right).

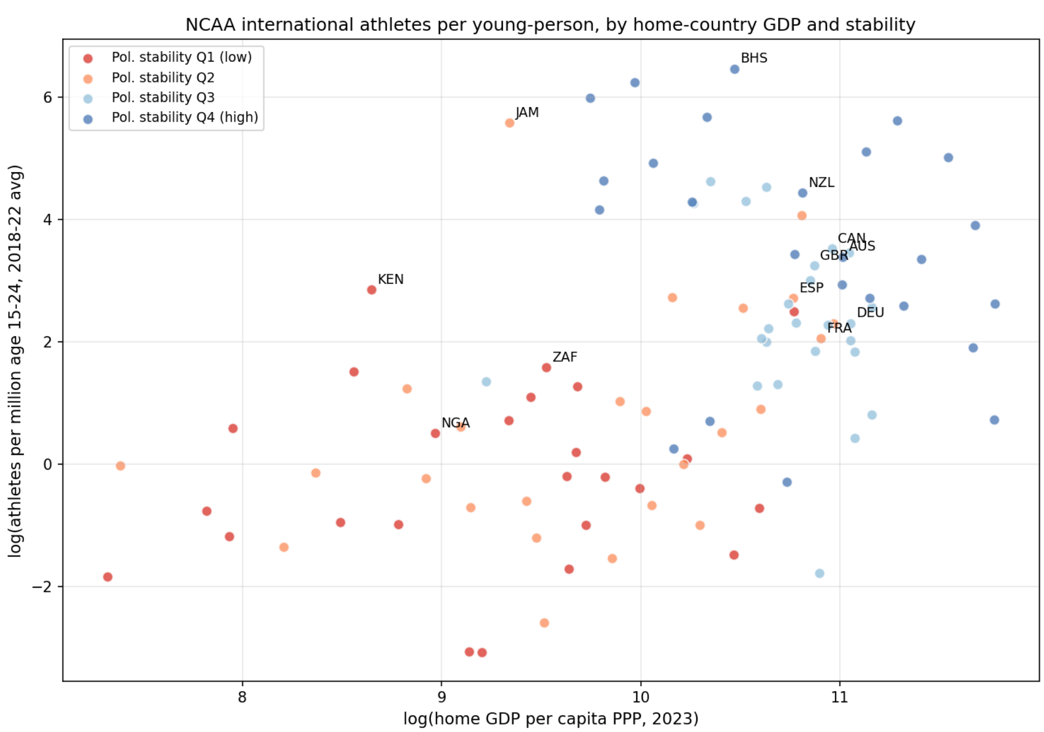

Figure 5: Log athletes per million age 15–24 vs. log home-country GDP per capita, colour-coded by political-stability quartile. The apparent stability gradient is largely a country-size artefact in this view; see Section 3.3.